

ABSTRACT

The Hospital Management Analysis project is a comprehensive healthcare analytics study aimed at transforming hospital operational data into actionable insights for improving patient care, resource utilization, and administrative efficiency. The analysis is based on an integrated Hospital Management dataset consisting of patient records, doctor information, appointment details, billing transactions, insurance claims, medicine inventory, laboratory reports, ward occupancy records, nurse details, and hospital information collected across multiple hospitals in Tamil Nadu between 2020 and 2025.

The project utilizes various data analytics tools and technologies, including Excel, SQL, Python, and Power BI, to perform data cleaning, preprocessing, analysis, and visualization. Python is used to handle missing values, detect inconsistencies, perform data transformations, and conduct exploratory data analysis. SQL facilitates efficient storage, querying, and retrieval of healthcare data from multiple relational tables. Excel supports preliminary analysis through formulas, PivotTables, and summary reports, while Power BI enables the creation of interactive dashboards for monitoring hospital performance and key healthcare indicators.

The analysis focuses on critical performance metrics such as patient admissions, appointment completion rates, department-wise patient distribution, doctor utilization, hospital capacity, ward occupancy, treatment costs, billing trends, insurance claim status, laboratory test activity, and medicine inventory management.

The findings provide valuable insights into hospital operations, helping healthcare administrators optimize staffing levels, improve bed utilization, streamline appointment scheduling, manage medical inventories effectively, and enhance patient service quality. Furthermore, the project demonstrates how data-driven decision-making can improve operational efficiency, reduce costs, and support strategic planning within healthcare institutions.

Overall, the Hospital Management Analysis project highlights the significance of healthcare analytics in modern hospital administration and showcases the effective integration of data management, statistical analysis, and business intelligence tools to improve healthcare delivery and organizational performance.